Tsinghua University
Department of Electronic Engineering
Probability and Stochastic Processes (1)
(Spring 2021)

Problem Set 7
Due: Apr. 12, 2021

1. Section 3.4 Problem 16.

   Consider the following variant of Buffon's needle problem (Example 3. 11), which was investigated by Laplace. A needle of length l is dropped on a plane surface that is partitioned in rectangles by horizontal lines that are a apart and vertical lines that are b apart. Suppose that the needle's length l satisfies $l < a$ and $l < b$. What is the expected number of rectangle sides crossed by the needle? What is the probability that the needle will cross at least one side of some rectangle?

   Solution: Let A be the event that the needle will cross a horizontal line, and let B be the probability that it will cross a vertical line. From the analysis of Example 3.11, we have that

$$P(A) = \frac{2l}{\pi a}, \qquad P(B) = \frac{2l}{\pi b}.$$

   Since at most one horizontal (or vertical) line can be crossed, the expected number of horizontal lines crossed is $P(A)$ [or $P(B)$, respectively]. Thus the expected number of crossed lines is

$$P(A) + P(B) = \frac{2l}{\pi a} + \frac{2l}{\pi b} = \frac{2l(a+b)}{\pi ab}.$$

   The probability that at least one line will be crossed is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

   Let X (or Y) be the distance from the needle's center to the nearest horizontal (or vertical) line. Let Θ be the angle formed by the needle's axis and the horizontal lines as in Example 3.11. We have

$$P(A \cap B) = P\left(X \le \frac{l \sin\Theta}{2}, Y \le \frac{l\cos\Theta}{2}\right).$$

   We model the triple (X, Y, Θ) as uniformly distributed over the set of all (x, y, θ) that satisfy $0 \le x \le a/2$, $0 \le y \le b/2$, and $0 \le \theta \le \pi/2$. Hence, within this set, we have

$$f_{X,Y,\Theta}(x, y, \theta) = \frac{8}{\pi ab}.$$

   The probability $P(A \cap B)$ is

$$\begin{aligned}
P(X \le (l/2)\sin\Theta, Y \le (l/2)\cos\Theta) &= \iint_{x \le (l/2)\sin\theta, y \le (l/2)\cos\theta} f_{X,Y,\Theta}(x,y,\theta)\,dxdyd\theta \\
&= \frac{8}{\pi ab}\int_0^{\pi/2}\int_0^{(l/2)\cos\theta}\int_0^{(l/2)\sin\theta} dxdyd\theta \\
&= \frac{2l^2}{\pi ab}\int_0^{\pi/2} \cos\theta\sin\theta d\theta \\
&= \frac{l^2}{\pi ab}.
\end{aligned}$$

Thus we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{2l}{\pi a} + \frac{2l}{\pi b} - \frac{l^2}{\pi ab} = \frac{l}{\pi ab}(2a + 2b - l).$$

2. Section 3.5 Problem 20.

   An absent-minded professor schedules two student appointments for the same time. The appointment durations are independent and exponentially distributed with mean thirty minutes. The first student arrives on time, but the second student arrives five minutes late. What is the expected value of the time between the arrival of the first student and the departure of the second student?

   Solution: The expected value in question is

$$\begin{aligned}\mathbb{E}[\text{Time}] =& (5 + \mathbb{E}[\text{stay of 2nd student}]) \cdot P(\text{1st stays no more than 5 minutes}) \\ &+ (\mathbb{E}[\text{stay of 1st}|\text{stay of 1st} \geq 5] + \mathbb{E}[\text{stay of 2nd}]) \\ &\cdot P(\text{1st stays more than 5 minutes}).\end{aligned}$$

   We have $\mathbb{E}[\text{stay of 2nd student}] = 30$, and, using the memorylessness property of the exponential distribution,

$$\mathbb{E}[\text{stay of 1st}|\text{stay of 1st} \geq 5] = 5 + \mathbb{E}[\text{stay of 1st}] = 35.$$

   Also

$$P(\text{1st stays no more than 5 minutes}) = 1 - e^{-5/30},$$
$$P(\text{1st stays more than 5 minutes}) = e^{-5/30}.$$

   By substitution we obtain

$$\mathbb{E}[\text{Time}] = (5 + 30) \cdot (1 - e^{-5/30}) + (35 + 30) \cdot e^{-5/30} = 35 + 30 \cdot e^{-5/30} = 60.394.$$

3. Section 3.6 Problem 34.

   A defective coin minting machine produces coins whose probability of heads is a random variable P with PDF

$$f_P(p) = \begin{cases} pe^p & p \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

   A coin produced by this machine is selected and tossed repeatedly, with successive tosses assumed independent.

   (a) Find the probability that a coin toss results in heads.

   (b) Given that a coin toss resulted in heads, find the conditional PDF of P.

(c) Given that the first coin toss resulted in heads, find the conditional probability of heads on the next toss.

Solution:

(a) Let A be the event that the first coin toss resulted in heads. To calculate the probability $P(A)$, we use the continuous version of the total probability theorem:

$$P(A) = \int_0^1 P(A|P=p)f_P(p)dp = \int_0^1 p^2 e^p dp,$$

which after some calculation yields

$$P(A) = e - 2.$$

(b) Using Bayes' rule,

$$\begin{aligned} f_{P|A}(p) &= \frac{P(A|P=p)f_P(p)}{P(A)} \\ &= \begin{cases} \frac{p^2 e^p}{e-2}, & 0 \le p \le 1, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

(c) Let B be the event that the second toss resulted in heads. We have

$$\begin{aligned} P(B|A) &= \int_0^1 P(B|P=p, A)f_{P|A}(p)dp \\ &= \int_0^1 P(B|P=p)f_{P|A}(p)dp \\ &= \frac{1}{e-2}\int_0^1 p^3 e^p dp. \end{aligned}$$

After some calculation, this yields

$$P(B|A) = \frac{1}{e-2}\cdot(6-2e) = \frac{0.564}{0.718} \approx 0.786.$$

4. Let Δ denote the event that you can form a triangle with three given parts of a rod R.

(a) R is broken in two uniformly at random, the longer part is broken in two uniformly at random. Show that $P(\Delta) = \log(4/e)$.

(b) R is broken in two uniformly at random, a randomly chosen part is broken into two equal parts. Show that $P(\Delta) = \frac{1}{2}$.

(c) In case (b) show that, given Δ, the triangle is obtuse with probability $3 - 2\sqrt{2}$.

Solution: We may assume without loss of generality that R has length 1. Note that Δ occurs if and only if the sum of any two parts exceeds the length of the third part.

(a) The length X of the shorter piece has density function $f_X(x) = 2$ for $0 \leq x \leq \frac{1}{2}$. The other pieces are of length $(1-X)Y$ and $(1-X)(1-Y)$, where Y is uniform on $(0,1)$. The event Δ occurs if and only if $2Y < 2XY + 1$ and $X + Y - XY > \frac{1}{2}$, and this has probability

$$2\int_0^{\frac{1}{2}} \left[\frac{1}{2(1-x)} - \frac{1-2x}{2(1-x)}\right] dx = \log(4/e).$$

(b) The three lengths are X, $\frac{1}{2}(1-X)$, $\frac{1}{2}(1-X)$, where X is uniform on $(0,1)$. The event Δ occurs if and only if $X \leq \frac{1}{2}$.

(c) This triangle is obtuse if and only if

$$\frac{\frac{1}{2}X}{\frac{1}{2}(1-X)} > \frac{1}{\sqrt{2}},$$

which is to say that $X > \sqrt{2} - 1$. Hence,

$$P(\text{obtuse}|\Delta) = \frac{P(\sqrt{2} - 1 < X < \frac{1}{2})}{X < \frac{1}{2}} = 3 - 2\sqrt{2}.$$